Towards an AI co-scientist

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Scientific discovery relies on scientists generating novel hypotheses that undergo rigorous experimental validation. To augment this process, we introduce an AI co-scientist, a multi-agent system built on Gemini 2.0. The AI co-scientist is intended to help uncover new, original knowledge and to formulate demonstrably novel research hypotheses and proposals, building upon prior evidence and aligned to scientist-provided research objectives and guidance. The system's design incorporates a generate, debate, and evolve approach to hypothesis generation, inspired by the scientific method and accelerated by scaling test-time compute. Key contributions include: (1) a multi-agent architecture with an asynchronous task execution framework for flexible compute scaling; (2) a tournament evolution process for self-improving hypotheses generation. Automated evaluations show continued benefits of test-time compute, improving hypothesis quality. While general purpose, we focus development and validation in three biomedical areas: drug repurposing, novel target discovery, and explaining mechanisms of bacterial evolution and anti-microbial resistance. For drug repurposing, the system proposes candidates with promising validation findings, including candidates for acute myeloid leukemia that show tumor inhibition *in vitro* at clinically applicable concentrations. For novel target discovery, the AI co-scientist proposed new epigenetic targets for liver fibrosis, validated by anti-fibrotic activity and liver cell regeneration in human hepatic organoids. Finally, the AI co-scientist recapitulated unpublished experimental results via a parallel in silico discovery of a novel gene transfer mechanism in bacterial evolution. These results, detailed in separate, co-timed reports, demonstrate the potential to augment biomedical and scientific discovery and usher an era of AI empowered scientists.

1 Introduction

Human ingenuity and creativity propel the advancement of fundamental research in science and medicine. However, researchers, particularly in biomedicine, are faced with a breadth and depth conundrum. The complexity of biomedical topics require increasingly deep and specific subject matter expertise, while leaps in insight may still arise from broad knowledge bridging across disciplines. With the rapid rise in scientific publications and the availability of numerous technologies for specialized high-throughput assays, mastery of both discipline-specific depth and trans-disciplinary insight can be challenging.

Despite these challenges, many modern breakthroughs have emerged from trans-disciplinary endeavours. Emmanuelle Charpentier and Jennifer Doudna won the 2020 Nobel Prize in Chemistry for their work on CRISPR [1], which combined techniques and strategies ranging from microbiology to genetics to molecular biology. These benefits of synergy have also been seen beyond experimental biomedicine in numerous other areas of science. Notably, Geoffrey Hinton and John Hopfield combined ideas from physics and neuroscience [2, 3] to develop artificial intelligence (AI) systems, which were awarded the 2024 Nobel Prize in Physics.

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